FORM HS-SD · REV. 5 · POST-STASIS REFRESHER · SUPPLEMENT D · POST AT EVERY MUSTER POINT & REST AREA

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

RESTED. LIT. FIT.

HEAVY WORK, POURS, LIFTS AND WELDING RUN AT NIGHT HERE. TIRED, IMPAIRED OR IN THE DARK, YOU CAN KILL A CREW.

JUST WOKE UP?

STASIS SCRAMBLED YOUR BODY CLOCK.

Supplement D (HS-SD) covers night work, fatigue and fitness for duty. Finish it before your first 100% shift. For the first 2–3 weeks after revival, most workers sleep in broken patches, wake at odd hours and get sleepy without warning. **You may not feel how tired you are.**

Old Earth research: 17–19 hours awake slows you about as much as a blood alcohol of 0.05%. 24 hours awake, about 0.10%.

- Tell **Revival Med Bay** how you're sleeping at your clearance check. They can adjust your first shifts.
- No sleeping pills unless a medic gives them. They can leave you drowsy on shift.
- Your heat ramp (HS-03) still applies at night whenever the level is Caution or higher or WBGT is 78+.

SHIFT LIMITS

EVERY STATION · EVERY SHIFT
YOUR CREW LEAD LOGS YOUR HOURS

12 h	10 h	3	8 h
MAX SHIFT	MIN OFF BETWEEN SHIFTS	NIGHTS IN A ROW, THEN 48 H OFF	MAX IN A SAFETY-CRITICAL ROLE

LIMIT	DCW LIMIT (SITE RULE)	OLD EARTH BASIS
Shift length	12 h max, from muster to the end of your handover. No overtime past 12 h. Only Hub Command can extend a shift, for a site emergency, and never past 16 h.	UK HSE HSG256: 12 h max including overtime. US NRC 10 CFR 26.205: 16 h max in 24 h.
Rest between shifts	10 h off minimum, from the end of handover to your next muster.	US NRC 10 CFR 26.205: 10 h break between work periods.
Breaks	A 30-minute meal break by hour 6, plus every rest break the heat work/rest table calls for.	HSE HSG256: regular breaks. US Army TB MED 507 work/rest.
Nights in a row	3 max, then at least 48 h off before your next shift.	HSE HSG256: limit long shifts and night shifts to 2–3 in a row.
Switching days and nights	At least 48 h off before you switch, either way.	HSE HSG256: two nights' full sleep before a switch.
Weekly	60 h max on shift in any 7 days.	US NRC 10 CFR 26.205: 72 h max in 7 days.

A schedule that breaks a limit is unsafe work. Say so to your crew lead. Every stop is logged and nobody is penalized for it.

SAFETY-CRITICAL ROLES

One slow reaction in these roles can kill someone else:

CRANE OPERATOR · QUALIFIED SIGNAL PERSON · LIFT DIRECTOR · ENTRY SUPERVISOR · ATTENDANT · FIRE WATCH · HEAVY EQUIPMENT & PRINTER OPERATORS · MEWP, FORKLIFT & TELEHANDLER OPERATORS · SHUTTLE & VEHICLE DRIVERS

- **Max 8 h in the role per shift, and never after hour 10 of your shift.** Hand it over to a rested worker with the same badge endorsement. (SITE RULE)
- **Short on sleep, no role.** Less than 5 h sleep in the last 24 h, or less than 12 h in the last 48 h: tell your crew lead at muster. You stay off safety-critical roles this shift. (SITE RULE)
- **Fire watch at its limit:** hand over face to face to a fresh fire watch and announce it on Ch 13. The PTW-HW stays open until the fire watch reports "watch complete".
- **Crane operators change only with the load landed.** (SITE RULE)

FATIGUE

REPORT IT · NO PENALTY

TIRED? SAY SO.

Tell your crew lead: **"I'm not fit for this task."** No penalty, no questions. It goes in the log as a fatigue report and never counts against you. (SITE RULE)

- Your crew lead moves you to non-critical work, gives you a rest break, or sends you off shift.
- Caught yourself drifting off? Make your work safe first, then call. At height: stay clipped in, sit down, call your buddy.
- Report anything that keeps cutting your sleep: a noisy barracks, long trips, a schedule that breaks the limits. Your crew lead takes it to the foreman.

SIGNS CREW LEADS LOOK FOR

- Yawning, heavy eyelids, rubbing eyes, head nodding.
- Staring, slow answers, missed radio calls or check-ins.
- Repeated mistakes, skipped steps, dropped tools.
- Short temper, shortcuts, poor judgment.
- **Microsleeps:** a few seconds asleep, often with the eyes open.

Seen one? Talk to them. Move them off safety-critical work and keep their buddy with them. Log it as a fatigue report.

Heat first. On a hot site, anyone who collapses or isn't fully alert is a heat-stroke case until a medic says otherwise. Ch 1: "Emergency, emergency, emergency. Heat casualty." Cool first, transport second (HS-03). Never write it off as tiredness.

SLEEP, REST & GETTING HOME

NIGHT CREWS SLEEP BY DAY

SLEEP

- Aim for 7 hours or more. Nap before your first night shift.
- Dark, cool and quiet. Night crews sleep in the barracks day-sleep wing, kept dark, cooled and quiet all day. (SITE RULE)
- Keep the same sleep times, even on days off when you can.
- Eat light before sleep. Screens off.
- Tinted glasses on the ride home keep morning light from waking your body clock.

REST AREAS & NAPS

- Every sector's large shelter has a quiet, cooled rest area, open all shift. (SITE RULE)
- On a break, you may nap up to 20 minutes in a rest area. (SITE RULE)
- After a nap, wait 15 minutes before you climb, drive or take a safety-critical role. The grogginess (sleep inertia) needs that long to pass.
- Never sleep in a vehicle or machine cab, under equipment, in a trench or on a deck.

CAFFEINE & STIMULANTS

- Coffee and tea are fine. Keep caffeine under about 400 mg a day: about 4 cups of brewed coffee.
- **No energy drinks or stimulant drinks on deck** (HS-03).
- No caffeine in the 6 hours before you plan to sleep.
- Caffeine doesn't replace sleep or water. Keep drinking water by the HS-03 rules.
- Stay-awake pills only from a medic, declared at your Forward Aid Station (FAS).

SHUTTLES & DRIVING

- Coming off a night shift? Ride the Transport shuttle. **Never drive a shuttle or site vehicle after your shift.** (SITE RULE)
- Shuttle drivers are Transport staff on their own rested shift. The safety-critical limits above apply to them. (SITE RULE)
- Drowsy at the wheel: pull over somewhere safe, park, and call for a relief driver. Never push on.
- Seat belt on, every ride.

SHIFT HANDOVER

EVERY SHIFT CHANGE · FACE TO FACE

1. **Meet at the work face.** Outgoing and incoming crew leads, in person. A radio call isn't a handover. Handover time counts inside your 12 h.
2. **Walk the permit board.** What's open and what closes now. A PTW-HW ends with its shift. Running fire watches hand over on Ch 13. A PTW-CS never runs past one shift: the new shift needs a new PTW-CS and fresh air tests. (SITE RULE) Entry supervisor or attendant changes mid-shift (8 h role limit): the incoming one signs the permit before the outgoing one leaves. Attendants pass the entrant count in person. (SITE RULE) Permits: REF-PTW.
3. **Locks.** The incoming worker's lock goes on before the outgoing worker's comes off. Walk what's isolated and what's still live (HS-07).
4. **Loads and machines.** Crane operators change only with the load landed. Pass on every defect and every machine tagged out.
5. **Conditions.** Heat level and WBGT, wind, warnings on Ch 16, open holes, removed guardrails, dark spots, open stop-works.
6. **People.** Who's on the ramp, on light duty or restricted, who reported fatigue, the buddy pairs, and every lone worker's logged location.
7. **Sign the handover log.** Both crew leads sign. The incoming crew lead briefs the crew at muster. (SITE RULE)

NIGHT LIGHTING

MINIMUM LIGHT AT THE WORK · FC = FOOT-CANDLES

OLD EARTH MINIMUMS

FC	WHERE (OSHA 1926.56, TABLE D-3)
5	General construction area lighting.
3	Excavation and waste areas, concrete placement, accessways, active storage, loading platforms, refueling and field maintenance areas.
5	Indoor corridors, warehouses and exitways. Tunnels, shafts and underground work (10 at a heading while drilling, mucking or scaling).
10	Shops and plants: Fabrication Yards, welding bays, equipment rooms, rigging lofts, barracks, mess halls, indoor toilets.
30	First aid stations (every FAS), infirmaries, offices.

1 fc is about 10.8 lux. Safety & Health spot-checks light levels each night shift (SITE RULE). Rule of thumb: if you can't read a sling tag or your permit where you stand, it's too dark to work there.

BRIGHTER WHERE IT COUNTS

FC	WHERE (SITE RULE)
10	Stairs, ladders and landings, floor-hole covers, leading edges and walkways from Deck 1 up.
10	Crane landing zones, rigging areas and the load path where people work, for the whole lift.
10	The 35 ft hot-work zone, including below it, so the fire watch can see where sparks land.
5	Haul-road crossings and walkways beside haul roads, with reflective posts.
Backup	Battery lights on every stair tower and exit route of an active structure. Emergency lighting in every shelter.

Confined spaces: work lights inside, a helmet light on every entrant, and intrinsically safe lights wherever flammables are possible (Supplement A (HS-SA)).

GLARE & HELMET LIGHTS

- Aim lights down and away from crane cabs, machine operators, haul roads and signal persons' faces.
- Light the signal person from the front. A backlit signal person is a silhouette, and the operator can't read the signals.
- Dazzled? Stop and call it in. The crew re-aims portable lights. Electrical & Power re-aims fixed ones.
- Every night worker carries a helmet light from the Quartermaster. It's a backup and never counts toward the minimums. (SITE RULE)
- Dip your helmet light when you face someone. Never shine it into a cab.
- Class 3 hi-vis at night.
- Light towers: outriggers out on mats. Mast down for a sandstorm warning or above the maker's wind limit.

LIGHT OUT? REPORT IT.

- Stop the work in the dark spot and keep people out of it.
- Tell your crew lead. The crew lead tells the foreman on the foreman talkgroup.
- The foreman calls Electrical & Power ("[Sector] GF Electrical") on the sector channel for the repair.
- Only qualified electrical workers (blue hats) repair, move or re-power fixed lighting and its cables. See Supplement B (HS-SB).
- You may move battery work lights.
- Loads moving or torches lit in an area that goes dark: anyone calls "ALL STOP, ALL STOP".

LIGHTS OUT AT HEIGHT

The lights fail on a deck, scaffold, MEWP or ladder. In the dark you can't see the edge.

**STOP.
HOLD.
CALL.**

Never feel your way along an edge or down an unlit stair or ladder. Wait for light.

1. **Stop where you are.** Don't walk, climb or step toward an edge.
2. **Check you're clipped in.** On a guarded deck with no anchor, crouch where you are and keep a hand on solid structure.
3. **Helmet light on.** Hold your position.
4. **Call your crew lead** on the foreman talkgroup. The crew lead counts heads by radio.
5. **Cranes:** the operator stops all motion and holds the load. It is landed only on the signal person's direction, once the landing zone is lit. **Hot work:** torch off. The fire watch keeps watching with a light.
6. **Move only** when the lights are back, or when your crew lead brings lighting and a route down.

FITNESS FOR DUTY

EVERY SHIFT · EVERY STATION

TOPIC	THE RULE
Alcohol	None within 24 h before any shift, and none on site. (SITE RULE) Alcohol dries you out and dulls the warning signs of heat illness.
Drugs	No recreational drugs on site or before a shift. Never come to work impaired. (SITE RULE)
Medicines that make you drowsy	Sleeping pills, strong painkillers, sedating allergy pills, muscle relaxants, some anxiety and mood medicines. Tell your FAS before your next shift. The medic decides what work you can do. Never stop a prescribed medicine to pass a check.
Medicines that raise heat risk	Water pills (diuretics), allergy pills (antihistamines), beta-blockers, anticholinergics, some antidepressants and antipsychotics, stimulants. Declare them to your FAS. They can change your ramp and your work/rest. See the heat field card (TOOL-HFC).
Privacy	Your crew lead hears only what work you can and can't do. The medic keeps the reason.
Illness	Fever, vomiting or diarrhea: don't work. Go to your FAS. It resets your heat ramp to Day 1 and your RAMP band goes back on.
Back to work	Off work for an illness or injury, or sent off shift by the FAS or Central Med: re-clearance at Revival Med Bay before your next shift. Heat exhaustion: done for the day, re-clearance before your next shift. Back in stasis for any reason: full revival clearance and Day 1 of the ramp.
Someone seems impaired	Slurring, stumbling or confused on a hot site: Ch 1 "Heat casualty", and cool them until a medic says otherwise. Fully alert, but you suspect drink, drugs or medicine: take them off the task, keep them off ladders, machines and vehicles, and walk them to the FAS. Never send them alone.

WELLBEING

AFTER STASIS · AFTER INCIDENTS
NO PENALTY FOR ASKING

NORMAL AFTER STASIS

Most revived workers feel some of this for weeks:

- Disorientation and memory gaps.
- Broken sleep and vivid dreams.
- Low mood, worry, short temper.
- Grief for lost time, and for people who didn't wake or stayed behind.
- Feeling cut off from your crew.

It usually eases. If it doesn't, or it gets worse, ask for help.

WHERE TO GET HELP

- **SRU counselors** at Revival Med Bay. Walk in, any shift. (SITE RULE)
- **Your FAS medic**, any time. They can refer you to Central Med counselors at the Hub.
- **Peer supporters:** trained crew members in every sector. Their names are posted at every muster point. (SITE RULE)
- It's private. Asking for help never costs you your badge. A medic may move you off safety-critical work for a while. That is care, and it's never logged as discipline.

WORRIED ABOUT SOMEONE?

- Signs: pulling away, not sleeping, drinking more, talking about being a burden or wanting to die, giving things away, wild risks at height.
- Ask them, privately and directly. Asking about suicide doesn't put the idea in their head.
- Listen. Stay with them and get your crew lead to take them to an SRU counselor or the FAS. A peer supporter or any medic can help too.
- **In danger right now?** Stay with them, keep them away from edges and machines, and call Ch 1: "Emergency, emergency, emergency. Man down", location, your call sign. Never broadcast the reason.
- Raising a concern never gets you or them in trouble.

AFTER A SERIOUS INCIDENT

- At the scene: get people safe, calm and back with their crew, and tell them what happens next.
- Everyone involved gets a medical and a debrief (HS-09), plus a check-in within 72 hours from an SRU counselor, Central Med or a peer supporter. Talking is offered. Nobody is forced. (SITE RULE)
- Crews directly involved stand down from safety-critical roles for the rest of that shift. (SITE RULE)
- Bad sleep, replaying it, jumpiness, anger or numbness are normal for a few weeks. Still bad after a month, or getting worse: see a counselor.

LONE WORK

- **None** at Caution or higher, at height, in confined spaces, or near wadis and trenches.
- Otherwise only with **15-minute check-ins** and a location your crew lead logs. Check in on your foreman talkgroup.
- Missed check-in: immediate call-back. No answer within 5 minutes: someone goes in person.
- No lone work in unlit areas at night. (SITE RULE)
- Nights can still be Caution or hotter. Check the posted level before you work alone.
- This covers survey crews, inspectors and anyone else who often works alone.

STOP-WORK AUTHORITY

Too tired, too dark, or someone impaired: that's unsafe work. **Anyone can stop unsafe work. No penalty, no questions.** Stop it, make it safe, tell your crew lead (your foreman if there's no crew lead; any green hat if it isn't fixed). Work restarts only when the crew lead confirms the fix with the person who stopped it. Every stop is logged.

WORKER ACKNOWLEDGEMENT

I know the shift and rest limits. I will report fatigue, impairment and medicines that affect my work, without fear of penalty. I know to stop and hold when the lights fail at height.

NAME ______________________
COLONIST ID ______________________
CREW / SECTOR ______________________
DATE ______________________

Adapted from Old Earth standards: OSHA 29 CFR 1926.56 illumination (Table D-3), UK HSE HSG256 Managing shift work, US NRC 10 CFR 26.205 work-hour controls, NIOSH shift work and long work hours guidance, AASM/SRS adult sleep guidance, US FDA caffeine guidance, CDC/NIOSH heat stress risk factors, ANSI/ISEA 107 high-visibility apparel, WHO Psychological First Aid. Hour limits, rest areas and lighting above Table D-3 are DCW site rules.

HS-SD · Dunewright Colony Works · Replace when faded